

[Question 1] (1 point)

(...)

Based on your understanding of biometric systems, what could the "Score" values displayed in debug mode represent? How would you systematically investigate and determine whether these values correspond to **similarities** or **distances**? Provide a detailed plan for your approach. Assume the biometric software is fully functional, allowing you to freely enroll, identify, and verify individuals in both regular and debug modes.

Score values represent either similarities or distances between a presented fingerprint and a corresponding sample the system found in the database. To determine whether the values are sim. or dist.:
1. enroll your own fingerprint 2. check the score using verification function. 3. Try to verify a different person ~~by~~ using your id. If the score became higher — it was distance. If the score became lower — it was similarity.

[Question 2] (1 point)

What risks arise from deploying that fingerprint recognition system in production while it remains in debug mode? If an attacker were to exploit this configuration, in what ways could they compromise or manipulate the system? Provide a detailed explanation.

The debug mode gives the user a lot of information about the system, allowing them to launch an attack more easily. By providing the user ID when presented with a print, you could be compromising user data if an attacker was using a latent print or a mold. Additionally, by displaying the scores, you run the risk of an attacker using a hill climbing attack, perfecting a fake fingerprint until it gave them access.

[Question 3] (1 point)

Given the system's "Score" values (representing either similarities or distances), how would you evaluate the overall performance of such a biometric solution? Specify the metrics you would report and the types of graphs you would generate to illustrate the results.

I would generate imposter and genuine pairs along with their scores. I would plot the distribution of these scores and calculate the D'Prime score (D') to measure how well-separated the genuine and imposter distributions are. A high D'Prime Score indicates good separation, meaning the system effectively distinguishes between valid and invalid users. I would also generate an AUC plot, which assesses the system's accuracy by maximizing the True Match Rate (TMR) and TNMR across various thresholds. A high AUC score reflects strong overall performance. If the system achieves both a high D'Prime and AUC, it would be considered reliable and effective. This approach provides a clear understanding of how well the system differentiates between legitimate and imposter attempts.

[Question 4] (1 point)

How would you use the "Score" values to control the trade-off between False Match Rate (FMR) and False Non-Match Rate (FNMR)? Given this particular application, which error would you prioritize reducing, and why? Explain how your choice would affect the other error rate.

I would use the score values to identify the best possible operating threshold of the biometric system. This can also include calculating the Equal Error Rate, where FMR & FNMR are equal. This starting point would allow us to adjust the operating threshold as we saw fit. For this application, I would prioritize reducing the FMR, as this would reduce the chances of someone being granted access to the building system, what have you; compared to falsely denying someone entry who actually belongs. This decision would increase the FNMR, as they are inversely proportional.

[Question 5] (1 point)

The hospital chain's managers have decided to acquire the fingerprint recognition solution. The discussion now involves (1) the need for presenting an identification card, along with the fingerprints, or (2) simply presenting the fingerprints and letting the system find who the person is. Which approach represents **biometric verification** and which represents **biometric identification**? Explain the fundamental difference between the two alternatives and discuss one advantage and one disadvantage of each approach for this application.

Approach 1) Biometric verification. The pros of this system would be that it reduces the computational load for the system, as well as adding an additional "verification" of having the ID. It may slow down the physical speed of the employee though.

Approach 2) Biometric identification. While the pros might be ease of use, the cons of having to use a feature gallery instead of template lowers the accuracy of the system and increases overhead.

[Question 6] (1 point)

The hospital chain's managers have finally decided to adopt a biometric verification approach. They will acquire a version of the system that utilizes a single-finger USB optical sensor with a resolution of 1200 ppi, along with an identification card reader. The complete specifications state that the provided software supports level-1, level-2, and even level-3 features. **Explain what each of these feature levels represents.** In the context of **biometric verification**, which feature level is the least useful, and why? Please justify your answer.

Level 1 is detection of the naked eye that gives the system classification. Level 2 is where the minutiae is calculated, this is when the system match and stores data. Level 3 is when more details of the fingerprint is done, with focus of sweat pores, ridges, and lifetime scars could be seen. This is more about liveness detection. Level 1 would be the least useful in the context of Biometric Verification, since, there is not a classification need for the program, the ID is presented, there is a focus with matching and liveness.

[Question 7] (1 point)

After adopting the fingerprint-based access control system, one hospital director proposed extending its use to record employee attendance and generate work-hour reports. What potential privacy, security, or ethical concerns could arise from this proposed use of the biometric system? Discuss and justify your answer.

Instances of "functional creep" will likely arise from this new implementation. Using biometric data to police employee ~~private~~ ~~status~~ behavior strays from the system's original boolean task of granting or denying access. Recording artifacts based on authentication may lead to the leakage of biometric data and/or PII, and may incentivize ~~ways~~ employees to figure out ways of tricking the system into recording false attendance... and therefore potentially leading to ~~unauthorized~~ unauthorized access.

[Question 8] (1 point)

A hospital manager is concerned that the fingerprint recognition system may quickly become obsolete, since the hospital managers expect to use it and its biometric database for at least the next ten years. As an alternative, they are also contemplating a more expensive face recognition system, which the manager believes is a more advanced technology. What factors should be considered when deciding whether fingerprint or face is the more appropriate biometric trait for this application? Based on these factors, which trait would you recommend for the hospital's access-control system? Justify your recommendation.

I recommend the hospital use fingerprints for access-control systems because if there is someone who intercepts / exfiltrate facial acquired data that is bad given the iris/facial textures cannot be subjected to change. Whereas, if an adversary attains fingerprint data, one can shift to another finger when system patches are in place. Fingerprints do provide more anonymity to employees in comparison to using facial traits. One's face could also change in 10 years whereas fingerprints remain the same.

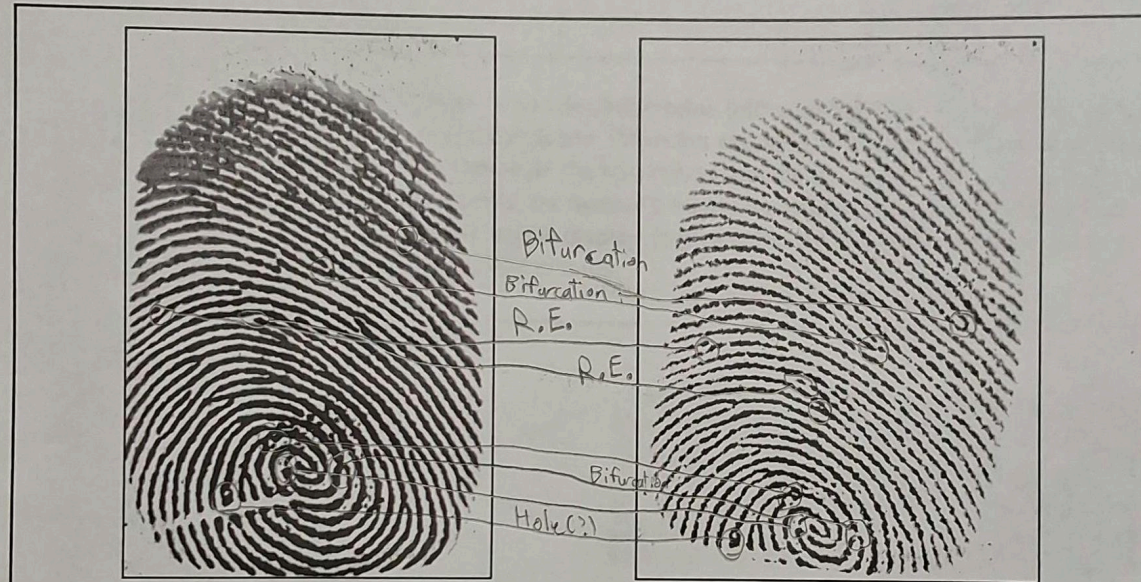
[Question 9] (1 point)

An actual case of a scientific paper submitted to a conference. While proposing a novel machine learning solution for fingerprint recognition, two authors designed an experimental setup in which they collected multiple fingerprint slaps from all the fingers of a large set of different people. To generate genuine and impostor pairs, they adopted the following approach: impostor pairs were generated by pairing individual finger slaps from different people, while genuine pairs were generated by pairing finger slaps from the same hand. With this configuration, they trained their solution and provided an ROC curve over the collected dataset. **Why was their paper a straightforward reject?** Please explain your answer.

If they were only making genuine pairs to the same hand and not the same finger then the genuine pairs metrically would look similar to the impostor, because if a person's thumb and index are compared they will look foreign to each other even though they are from the same hand.

[Question 10] (1 point)

Do the two fingerprints below depict the same finger? Please justify your answer by marking, linking, and type-naming five or more minutiae between them. After you've done this process manually, please explain why it is useful and important to program computers to do the same task.



Computers are far more capable & extremely fast at classifying minutiae & determining a match (high throughput, repeatability) than a human could hope to be. It's also easy to understand & quantify results for a computer.

Yes, the 2 fingerprints are a match, judging by similar minutiae in similar positions as well as the loop (whorl?) near the bottom.